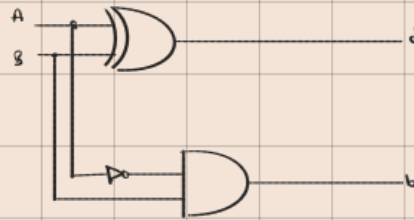


## Half Subtractor

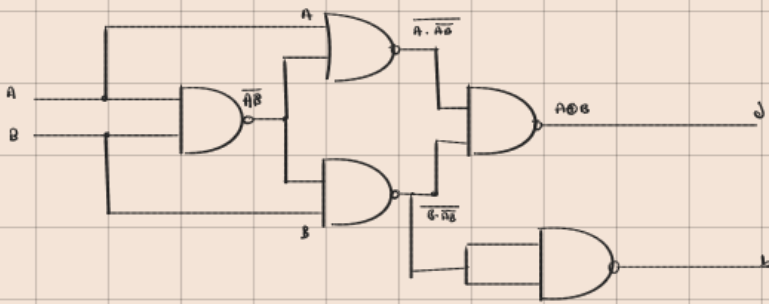
| A | B | d | b |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |

$$d = \bar{A}B + A\bar{B}$$

$$b = \bar{A}B$$



## Using Nand

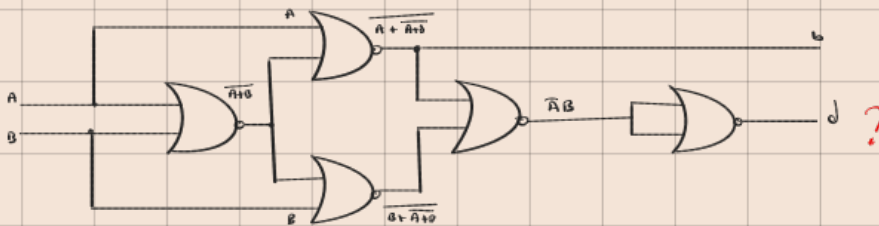


$$\overline{B \cdot A\bar{B}} \quad , \quad \overline{B \cdot A\bar{B}}$$

$$\overline{B \cdot A\bar{B}} + \overline{B \cdot A\bar{B}}$$

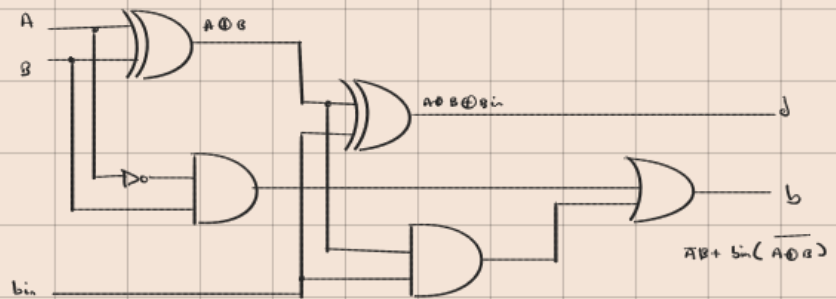
$$d\bar{B} + \bar{d}B + \bar{B}A$$

## Using Nor



Full Subtractor.

| A | B | $b_1$ | d | b |
|---|---|-------|---|---|
| 0 | 0 | 0     | 0 | 0 |
| 0 | 0 | 1     | 1 | 1 |
| 0 | 1 | 0     | 1 | 1 |
| 0 | 1 | 1     | 0 | 1 |
| 1 | 0 | 0     | 1 | 0 |
| 1 | 0 | 1     | 0 | 0 |
| 1 | 1 | 0     | 0 | 0 |
| 1 | 1 | 1     | 1 | 1 |



NAND Logic

I HATE TM'S

so I'm skipping this  
this khwani

NOR Logic.